UNIT 3

AEROSPACE PROPULSION SYSTEMS

Aircraft Inlets

An inlet or diffuser or an intake duct is a relatively simple duct with an increasing cross section along its length. The main function of an inlet is to decelerate the flow from high supersonic velocity to the velocity acceptable by the compressor (M=0.3 to 0.5).

CLASSIFICATION OF AIRCRAFT INLETS

Based on speed

1. Subsonic Inlets
2. Transonic Inlets
3. Supersonic Inlets
4. Hypersonic Inlets

SUPERSONIC INLETS ARE FURTHER CLASSIFIED AS FOLLOWS:

Based on method of compression

1. Normal Shock Inlets
2. Internal Compression Inlets
3. External Compression Inlets.
4. Mixed Compression Inlets

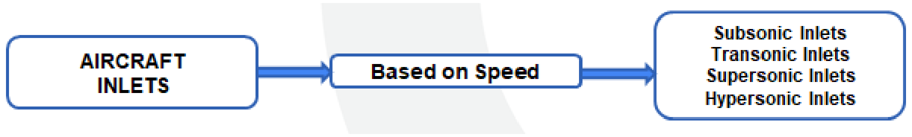
BASED ON GEOMETRY

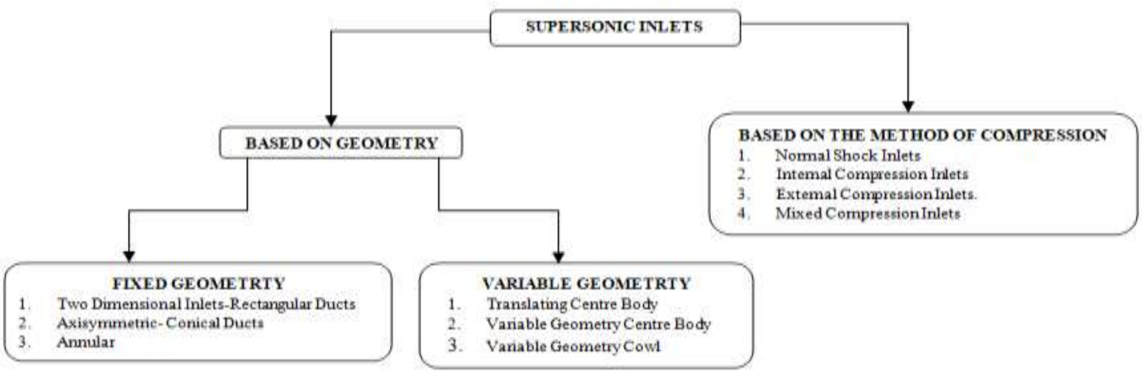
Fixed Geometry Inlets

1. Two Dimensional Inlets-Rectangular Ducts
2. Axisymmetric- Conical Ducts
3. Annular

Variable Geometry Inlets

1. Translating Centre Body
2. Variable Geometry Centre Body
3. Variable Geometry Cowl





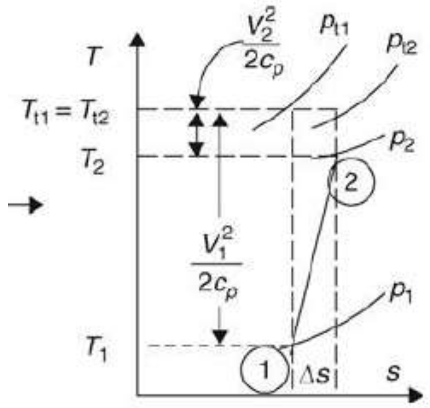
Subsonic Inlets/Pitot/Podded Inlets

Subsonic inlets are used in aircrafts whose speed is limited to high subsonic Mach numbers (Within M=1). The most common type of subsonic inlet is the pitot inlet as in the figure. Pitot inlets are most commonly used on Civil and military transport aircrafts. Pitot inlets have a fixed geometry with a diverging cross sectional shape along its length. The leading edge of the inlet is referred to as the lip. Usually a subsonic aircraft has an inlet with a relatively thick lip. The internal surface of the inlet is aerodynamically smoothened practically which offers zero resistance for the incoming air. As air flows into divergent duct, it slows and converts some of its kinetic energy into pressure in accordance with Bernoulli's theorem. This type of intakes makes the fullest use of ram effect due to forward speed. However, as sonic speed is approached, the efficiency of this type of air intake begins to fall because of the formation of a shock wave at the intake lip. Though subsonic inlets have fixed geometry, sometimes they are designed with additional blow in doors. We know that the highest thrust is needed during the take-off and climb phases where the aircraft speed is low. In order to generate maximum thrust, additional air flow is required which is achieved by opening the blow in doors. These doors are spring-loaded parts installed at the circumference of the inlet duct.

The function of the diffuser or inlet is to transform kinetic energy into a pressure rise, it is necessary to evaluate the total energy in the free stream and express it in terms of the total pressure that might be available for an ideal diffuser. The total pressure rise in a diffuser is given by the expression

$$\frac{P_{T0}}{P_0} = \left[1 + \frac{\gamma - 1}{2} M_0{}^2\right]^{\frac{\gamma-1}{\gamma}}$$

The T-s (Temperature-Entropy) diagram illustrates the thermodynamic process occurring in a diffuser



the performance of a diffuser will always be given by a parameter called as **Static Pressure Recovery Coefficient given as CPR**. Static Pressure Recovery Coefficient CPR is a non-dimensional parameter which indicates how well a diffuser has converted the kinetic energy of the free stream air into static pressure rise in the diffuser.

$$\text{CPR}=\frac{P_2 - P_1}{q_1}$$

Transonic Ducts

For operations above subsonic and below supersonic, transonic inlets are preferred. At transonic speeds (near Mach 1), the inlet duct is usually designed to keep the shock waves out of the duct. This is done by locating the inlet duct behind a spike or probe so that at airspeeds slightly above Mach 1.0 the spike will establish a normal shock bow wave in front of the inlet duct. This normal shock wave will produce a pressure rise and a velocity decrease to subsonic velocities before the air strikes the actual inlet duct. The inlet will then be a subsonic design behind a normal shock front. At low supersonic Mach numbers, the strength of the normal shock wave is not too great, and this type of inlet is quite practical. But at higher Mach numbers, the single, normal shock wave is very strong and causes a great reduction in the total pressure recovered by the duct and an excessive air temperature rise inside the duct.

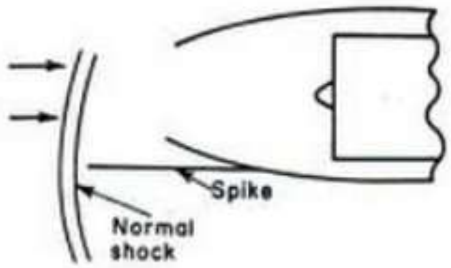


Figure: Schematic of a transonic duct with an upstream standing normal shock wave

Supersonic Inlets

The function of a supersonic inlet is the same as the function of a subsonic inlet, namely, to decelerate the flow to the engine face typically in the range of M2 ∼ 0.4–0.6, efficiently. Supersonic diffusers aid in decelerating the flow from supersonic to subsonic conditions. The deceleration of fluid from supersonic to subsonic velocities involves the formation of shock waves and in practice the study of supersonic inlets is very much dominated by the study of shocks intersecting, interacting, reflecting, and the shock boundary layer interaction.

Classification based on method of compression

(1) Normal shock inlets

Normal shock inlets are the simplest type of supersonic diffuser which utilizes a single normal shock wave to achieve the required static pressure rise by decelerating the flow. Normal shock inlets are very similar to the pitot inlets used on subsonic aircrafts. They have a simple geometry with a diverging cross section. But the only difference is that the lip of the normal shock inlet is made sharper than the subsonic inlet. The sharp lip geometry allows for an attached shock, whereas a blunt cowl leading edge creates a bow shock with the attendant external drag penalty. This type of supersonic inlet is short, light weight, and with no movable surfaces, which is suitable for low supersonic Mach applications or low-cost weapon systems development.

This type of duct is designed to operate with a normal shock wave attached to the lip of the inlet. When an aircraft with a normal shock inlet approaches the speed of sound, the sharp lip of the inlet establishes a single strong normal (perpendicular) shock wave at the entry plane of the inlet as in fig. As the flow passes through the shock wave, a reduction in velocity occurs with simultaneous increase in the static pressure rise, thus achieving subsonic velocities, which is further supplied at the face of the compressor.

The normal shock inlet operates in three different modes.

1. Critical mode
2. Sub-critical mode
3. Super-critical mode

The mode of operation of a normal shock inlet is dependent on the back pressure Pb maintained at the entrance of the compressor or the exit of the diffuser. Based on the back pressure, the location of the normal shock waves varies.

During the operation, the position of the normal shock wave depends on the inlet backpressure, established by the engine. The best backpressure, corresponding to design Mach number, places the normal shock at the lip where the best total pressure recovery coincides with zero spillage drag. This mode of operation is called the critical mode.

In the event of higher backpressure, that is, when the engine mass flow rate drops, the shock stands outside the inlet and a spillage flow takes place. This is the so-called subcritical mode of operation. The shock is drawn into the inlet, beyond the lip, when the engine backpressure is lowered. This is the so-called supercritical mode of operation. In the supercritical mode, the shock Mach number is higher than M_0, hence a larger total pressure drop in the inlet results. Consequently, the corrected mass flow rate at the engine face increases, which results in an increase in axial Mach number, thereby, reducing the engine face static pressure, that is, the inlet backpressure. Furthermore, the shock inside the duct may interact adversely with the wall boundary layer and cause separation and increase engine face distortion.

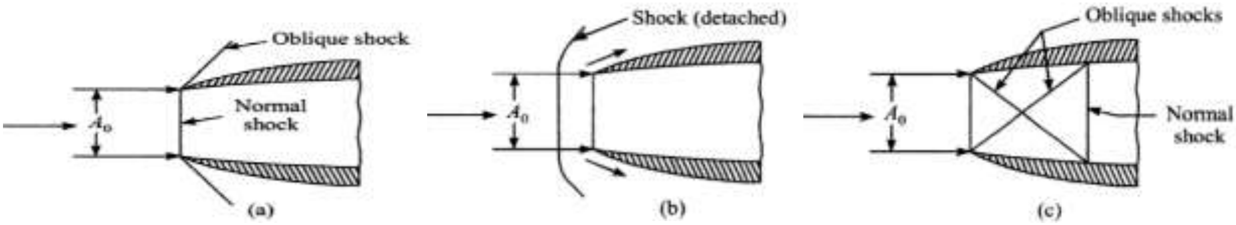
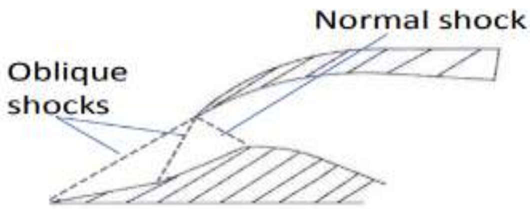


Figure: Various modes of operation of a normal shock inlet (a) critical mode (b) sub-critical mode (c) super-critical mode

(2) External Compression Inlets

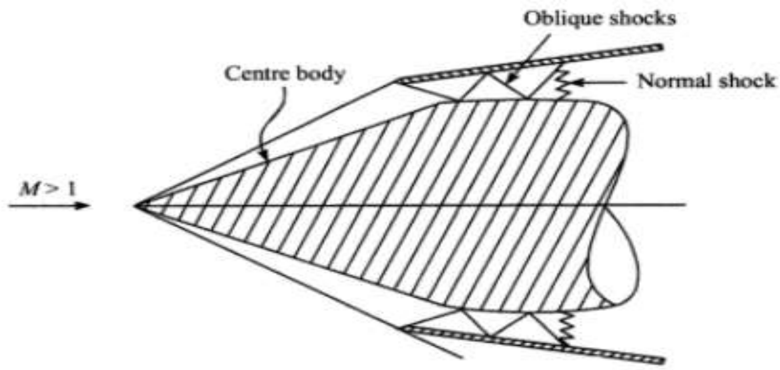
At low supersonic Mach numbers, the strength of the normal shock wave is not too great, and normal shock inlet is quite practical. But at higher Mach numbers, the single, normal shock wave is very strong and causes a great reduction in the total pressure recovered by the duct and an excessive air temperature rise inside the duct. Beyond this Mach number, the normal shock deceleration becomes inefficient, as it creates excessive total pressure loss. To extend the operational Mach numbers to ~2.0, a series of oblique shocks and normal shock wave are utilized which can decelerate the flow and provide sufficient pressure recovery. This can be done using external and internal compression inlets.



In an external-compression supersonic inlet, when the aircraft is operating at supersonic speeds, oblique and normal shock waves will be established. The oblique shock wave will occur between the forebody and the intake lip outside the inlet geometry and the normal shock wave will be established at the cowl lip which acts as the throat section as shown in the fig. During working, first the supersonic flow passes through the oblique shock waves which aids in the process of supersonic diffusion i.e., deceleration from high supersonic to low supersonic. Further, as the flow passes through the normal shock wave at the cowl lip, the flow is subsequently decelerated form low supersonic velocity to low subsonic velocity required by the compressor. In external compression inlet, the supersonic diffusion is completed entirely outside the cowl lip and hence the name.

(3) Internal Compression Inlets

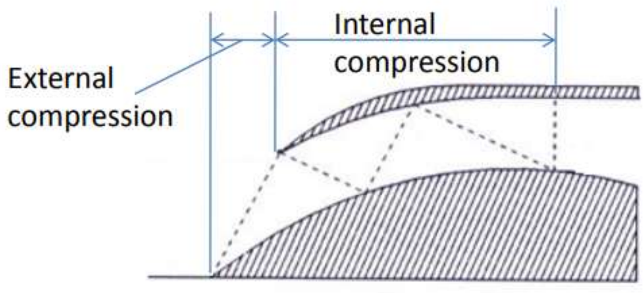
In internal compression inlets, the oblique & normal shocks are present within the covered passage of the inlet. Internal compression inlets are designed in such a way that the oblique shocks established within the inlet geometry undergo multiple reflections from the wall and terminate with a normal shock wave at the throat section existing downstream to the cowl lip. Here, due to the geometrical configuration of the inlet, both, oblique and normal shock waves occur within the inlet as shown in the figure. At high Mach numbers, series of oblique shock waves will be established and culminated in a terminal shock wave (normal S W) at the throat section of the inlet.



During operation, the supersonic flow passes through a series of oblique shock wave which helps in supersonic diffusion and achieve the deceleration of the fluid from high supersonic to low supersonic regimes. Later, the low supersonic flow passes through the normal shock wave established at the throat of the inlet and undergoes subsonic diffusion where in the flow is converted from low supersonic conditions to low subsonic condition. Due to the combination of multiple oblique shock waves, this type of inlet is suitable for speeds beyond Mach 1.5. By increasing the number of oblique shock waves, higher total pressure recovery can be achieved and further it can be used for higher Mach numbers.

(4) Mixed Compression Inlet

Mixed compression inlets have shocks that are located within as well as outside the intake geometry. Mixed compression inlets combine the features of both external as well as internal compression inlets and appear to be the most attractive design for most supersonic applications. The inlets utilizing a pattern of three or more shocks are frequently called mixed compression or external-internal contraction inlets. These inlets capitalize on the better total pressure recovery ratios available with internal shock reflections oblique shock waves. The oblique shock is reflected from the opposite wall, and the flow passing the reflected shock is restored to an axial direction. However, mixed compression inlets are susceptible to starting difficulties and may expel the normal shock (or "unstart") if the inlet is operated too far from the design conditions. Consequently, variable geometry is frequently used with mixed compression inlets. The operational Mach number of a mixed compression inlet extends to Mach ~ 4.0–5.0.



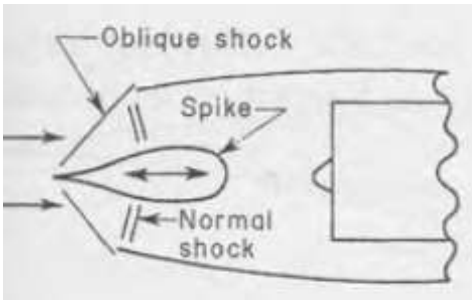
Variable Geometry Inlets

An aircraft equipped with a supersonic engine has to operate in all three speed regimes viz. subsonic, transonic and supersonic flows. Each of these speed zones requires a different inlet duct design. Since it's impossible to have three different inlets on an aircraft, the design should be such that the diffuser should be capable of handling all three regimes efficiently. One more problem is that when the aircraft with a supersonic duct flies at or near the speed of sound, shock waves will be established. If the shock waves are not controlled, it will give high duct loss and will set up vibrating conditions in the inlet duct, called inlet buzz. Buzz is an airflow instability caused by the shock waves rapidly being alternately swallowed and expelled at the inlet of the duct.

In order to accommodate the above conditions and also to satisfy the demands of the engine, the construction of the supersonic diffuser is usually incorporated with fixed and variable geometry inlets. Classification of supersonic inlets based on geometry is given below.
Fixed Geometry Inlets

Axisymmetric duct (Moving the inlet spike in and out so as to maintain the oblique shock on the edge of the outer lip of the duct**)**

All axisymmetric inlets are equipped with a conical forebody as shown in the figure. The centre forebody will be capable of moving forward and rearward which aids in opening and closing the throat area of the duct.



Two Dimensional Inlets-Rectangular Ramps (Moving the side wall or ramp to a higher angle so as to force a stronger oblique shock front**)**

We know that at different Mach numbers, altitudes within the flight envelope engine mass flow rate demands are different. For this reason, external compression ramps or solid wedges are incorporated on the inlets. These external ramps are hydraulically controlled which control their position during various flight phases. By changing the ramp angle, the ramp can be either extended into the flow or withdrawn from the flow depending on the engine requirements. The withdrawn position, which is also referred to as collapsed ramp position, allows for a larger inlet area needed for the take-off, climb, subsonic cruise, and transonic acceleration.

Compressors

Compressor is one of the important components in any gas turbine engine. The role of a compressor in any gas turbine engine is to supply high pressure air which will be heated in the limited volume of the combustion chamber and subsequently expanded in the turbine to generate work. The basic requirements of an efficient compressor are

1. High mass flow rate per unit frontal area of the engine

2. Provide with adequate mass flow rate throughout the flight envelope of an aircraft
3. High pressure ratio per stage
4. High efficiency
5. Possibility of multi staging

To perform the above said functions, there are two different types of compressors used on an aircraft jet engine and they are

- Centrifugal Compressors
- Axial Flow Compressors

Centrifugal Compressors

Like any other compressors, a centrifugal compressor is a prime mover mainly used to increase the pressure of the working fluid. In a centrifugal compressor, air enters the compressor axially and exits radially. In this type of compressor, the energy is dynamically transferred from the rotating member to the working fluid and the fluid is continuously supplied at a given pressure. The essential parts of a centrifugal compressor are as given below:

1. **Inlet casing with an accelerating nozzle -** Its function is to accelerate the working fluid to the impeller.
2. **Impeller -** It is the only rotating member of the compressor which helps in dynamically transferring the energy from the compressor to the working fluid.
3. **Diffuser -** It is the static component of the compressor used to help in the process of energy transformation.
4. **Volute -** Facilitates in collecting and channelizing the pressurised fluid into the duct or the next component of the system.

During the working of a centrifugal compressor, as the impeller rotates, initially a small negative pressure difference will be created at the entrance of the inlet casing. Due to this negative pressure, the air form the atmosphere will be drawn into the inlet casing. Since, the inlet casing is converging in cross section, as the air passes through the inlet, the pressure energy of the fluid will be converted into kinetic energy, thus increasing the velocity of the air. The increase in velocity at the expense of pressure is given by the curve (0-1). This high velocity air will be next passed onto the impeller. Due to the high rotational speeds of the impeller, large centrifugal forces will be established which will cause the air to move from the axis of rotation towards the outer periphery of the impeller. The passages of the impeller are diverging in cross section and as the air passes through the impeller, the kinetic energy of the air will be partially converted to the static pressure rise and illustrated by the curve (1-2). Now, this slightly high pressure air will be further supplied to the next component, the diffuser. In the diffuser all the kinetic energy achieved in the inlet casing will be completely converted into static pressure rise resulting in a high pressure working fluid. This increase in pressure is given by curve (2-3).

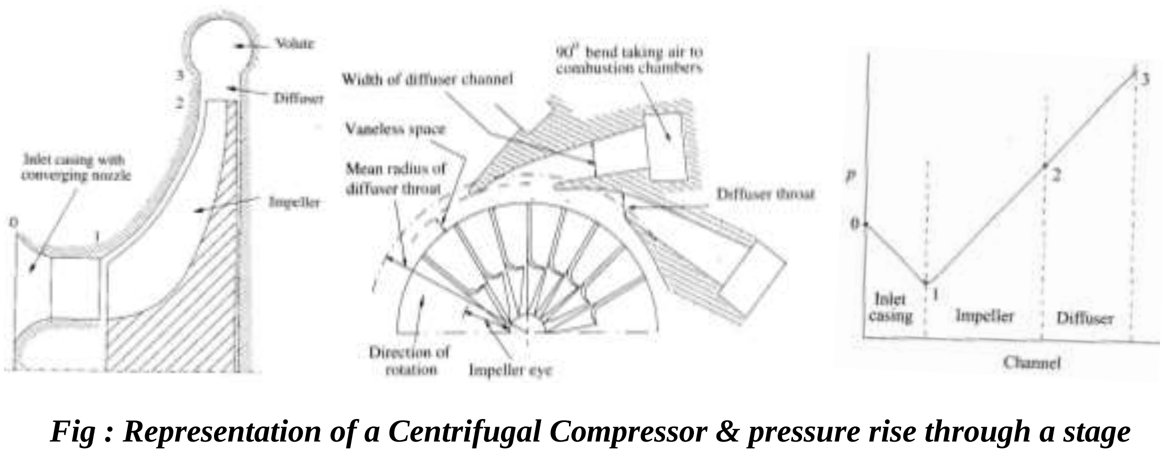


Fig : Representation of a Centrifugal Compressor & pressure rise through a stage

In a centrifugal compressor, 50% rise in static pressure is achieved in the impeller and remaining 50% rise is achieved in the diffuser. Thus the process of energy transfer and energy transformation occurs in the impeller and the diffuser respectively.

Advantages

1. Relatively simple in construction
2. Rugged and sturdy
3. Less susceptible to foreign object damage (FOD)
4. Performs well in impure environment
5. High compression per stage (Around 5 per stage)
6. Flexibility of operation is high

Disadvantages

1. Large frontal area, high drag
2. Multistaging is difficult
3. Low mass flow rate per unit frontal area
4. Efficiency is lower compared to axial flow compressors
5. Suitable only for smaller engines
6. When the pressure ratio is increased beyond 5, the efficiency drops rapidly due to the formation of shocks.

Axial Flow Compressors

Similar to the centrifugal compressors, even the axial flow compressors are prime movers used to increase the pressure of the working fluid by using energy form an external source. Major difference between centrifugal and axial flow compressor is that in centrifugal compressors, the flow is directed perpendicular to the axis of rotation but whereas in the axial flow compressors, the flow remain parallel from entrance to exit.

An axial flow compressor consists of an alternating sequence of rotating and stationery set of blades. The blades which are fixed to the outer casing and stationary are called as stator blades whereas the blades which are fixed to the spindle of the connecting shaft and rotating in nature are called as rotor blades. The rotor and the stator blades are arranged as close as possible for smooth and efficient flow with minimal pressure losses. One set of stator and rotor blades constitute a stage and the number of stages depends upon the pressure ratio required to be achieved in the compressor.

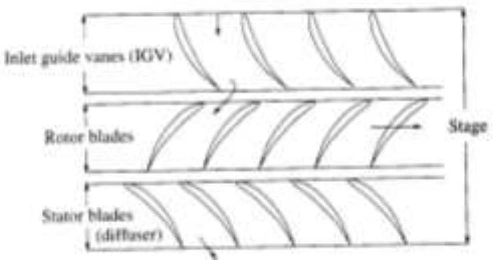


Fig : Axial Flow Compressor Stage

The axial flow compressor consists of a set of rotating blades followed by a set of stationary blades. During the working of an axial flow compressor, the air from the inlet duct will be passed to the rotor through the inlet guide vanes. The function of an inlet guide vane (IGV) is to direct the fluid to the rotor at a suitable angle, so that the fluid smoothly enters the rotary blades without any pressure losses. As the fluid passes into the compressor, first the rotor blades exert a torque on the fluid. During the interaction of the air with the rotating blade, the kinetic energy from the rotating blade will be transferred to the working fluid thus causing the fluid to accelerate. Further, this high velocity fluid will be then passed through the stator. Since the stator blades have a diffusing passage, as the fluid passes, the kinetic energy of the fluid will be converted into static pressure rise in the stator blades. In this way, the pressure of the working fluid will be increased with the passage through the stages of the compressor. In an axial flow compressor, the rotor blades help in the process of energy transfer and the stator blades help in the process of energy transfer.

Advantages

1. High efficiency
2. High mass flow rate per unit frontal area
3. Small frontal area
4. Highly suited for Multistaging

Disadvantages

1. Stage pressure ratio is less compared to centrifugal compressors (Around 1.15-1.6)
2. Susceptible to foreign object damage (FOD)
3. Effect of deposits have an adverse effect on the operation of the compressor
4. Delivery pressure is less

Technically, both, centrifugal and axial flow compressors are rotating machines (prime movers) and they are generally referred to as turbomachines. A turbomachine can be defined as a device in which energy transfer occurs between the working fluid and the rotating elements due to the dynamic action of the rotating elements, which results in the change of pressure and momentum of the fluid. This means that there will be transfer of mechanical energy into or out of the turbomachine depending on the type of turbomachine.

Combustion Chamber

Introduction:

Most advanced aircraft gas turbine engines usually employ two combustion chambers namely, main burner and the afterburner. In both the cases, liquid fuel is burnt releasing a large amount of heat energy. The combustion process is of critical importance in any gas turbine engine. This is because, in this process, the chemical energy of the fuel is converted into heat energy, which is later reconverted into useful work by the turbine. So, in any case of losses in the combustion chamber, it will have a direct effect on the thermal efficiency of the cycle.

The combustion chamber provides housing for the combustion process to occur. The process of combustion in gas turbine combustion involves the following important steps:

1. Formation of reactive mixture
2. Ignition
3. Flame propagation
4. Cooling of combustion products with air

A typical gas turbine combustion chamber consists of an invariably straight cylindrical duct which is incorporated with a central liner or flame tube equipped with a baffle plate. The flame tube will be usually placed in the centre which provides a small annular spacing between the outer casing and the flame tube itself. The main function of the annular spacing is to provide cooling air stream which reduces the temperature of the flame tube and the casing.

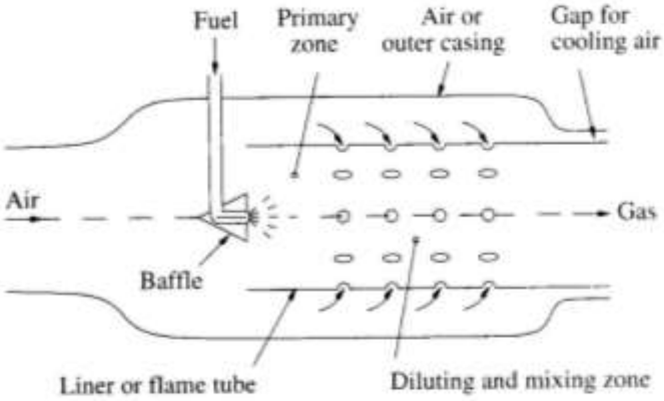


Fig: Illustration of a Combustion Chamber

During the working of a combustion chamber (both burner and afterburner), first the fuel will be introduced at the front end of the burner. The bulk fuel supplied will be either in a highly atomized spray from specially designed nozzles or in a vaporized form from devices called vaporizing tubes. If a nozzle is used then the fuel supplied will be in a highly atomized form of if a vaporizing tube is used then the fuel supplied will be in the form of vapours. Along with fuel, the high pressure air from the compressor will be also supplied to the combustion chamber. The fuel in the form of fine spray or vapour mixes with the incoming air and form a combustible mixture before the actual reaction takes place. On ignition, flame is established in the combustion chamber, raising the temperature and thus making combustion self-sustained.

Normally, the Stoichiometric mixture strength for most of the hydrocarbon fuel is 15:1. But usually in a gas turbine engine an overall air-fuel ratio of 60:1 to 100:1 is supplied. During combustion only one fourth of the total air will be utilized to achieve combustion and the rest of the air is used for cooling purposes. In the combustion chamber, the air supplied will be supplied in proportions at three different zones:

a) **Primary Zone:** In the primary zone about 15-20% of the air will be introduced around the fuel nozzle. The primary zone is provided with flame stabilizers like baffle plates which help in establishing a recirculation zone. Along with baffles a large number of small holes will also be provided along the liner which admits the air in the form of jets. This air will move upstream against the direction of fuel supply and causes turbulence. This turbulence helps in creating a thorough mixture between the air and fuel and resulting in a combustible mixture. Finally, necessary combustion will be provided through the Igniter plugs located in the primary zone.
b) **Secondary or Intermediate Zone:** In this zone around 30% of the total air will be introduced in the secondary zone which helps to achieve complete combustion. There might be some regions in the combustion chamber where the fuel wouldn't have mixed with air and is still in unburnt state. So when secondary air is introduced they mix up with the unburnt fuel and helps in completing the combustion process. The sum of primary and secondary air taken as total air-fuel ratio.
c) **Tertiary or Dilution zone:** The remaining air coming from the compressor is mixed with the products of combustion to cool them to the temperature required at the turbine inlet. Sufficient turbulence must be provided so that the hot and cold streams are thoroughly mixed to give a desired outlet temperature distribution, with no heat streaks

present in the exhaust gases. Presence of heat streaks or heat zones in the exhaust gases will damage the turbine blades.

The gases that result from combustion will have temperature close to 2000 degrees. Usually the liner walls are exposed to high temperatures. To protect the walls, cooling air is introduced at several stations along the liner thereby forming an insulating blanket between the hot gases and metal walls.

Classification of Combustion Chambers/ Burners

Types of combustion chambers or Burners

- Can Type
- Annular Type
- Can-Annular Type

Can Type Burners

- In can type, individual burners or cans are mounted in a circular form around the engine axis as shown in the fig.
- Here each can acts as a separate entity and will consist of separate injector, Swirler, atomizer, separate fuel injectors and receives air supply through its own cylindrical shroud.
- Replacement of damaged components is very easy.
- Its quite easy to control air-fuel ratio
- Can type combustors were used in olden day aircrafts and they are seldom used in newer aircrafts.
- It is difficult to achieve a uniform temperature distribution at its exit.
- Pressure drop is around 7% which is quite high.
- These type of configuration are usually found in combustion chamber incorporating centrifugal compressors
- Can type are mainly preferred in APUs rather than large engines

Advantages

1. Burners can be individually removed for inspection
2. Air fuel patterns can be easily controlled

Disadvantages

1. It occupies large space resulting in an increased size of the engine.

Annular Burner

- The annular burner consists of a single flame tube with a concentric cylinder mounted co-axially about the engine axis.
- Annular combustor is placed connecting directly the pressure delivery of the compressor and turbine inlet.
- In annular combustors, air from the compressor enters the annular diffuser at the entrance of the combustor. Fuel will be injected by series of injectors.

- This type of burner comes with large number of holes in the combustion liner and is mainly used for cooling purposes.
- Such type of burners is mainly suited for axial flow compressors.

Advantages

1. It has improved exit temperature distribution
2. Has increased durability
3. They have less surface to volume ratio, which requires less air for cooling
4. Improved performance
5. Weight is less
6. Pressure losses are less when compared to other two types (5%)
7. Has highest efficiency

Disadvantages

1. Suffers from structural problems due to large diameter thin walled cylinder. This problem is more severe in large engines.
2. Structurally weak
3. During inspection and repair, the entire combustor has to be removed.

Can-Annular Burner

- This type of burner combines the configuration of both can and annular designs.
- It has individual flame tubes arranged uniformly in the annular casing
- This design makes good use of available space.
- It employs a number of individually replaceable cylindrical liners that receives air through a common annular housing and also provides good control and air flow patterns.

Advantages

1. Greater structural stability
2. Less pressure losses
3. Less space requirements

Disadvantages

1. Expensive to repair
2. Non uniform temperature distribution
3. Thermal expansion of liners

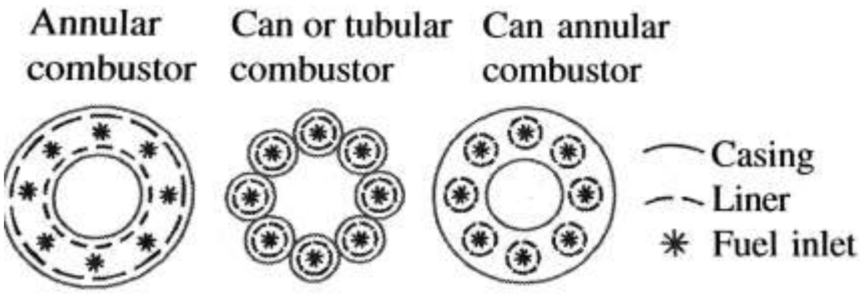


Fig: Types of burners

Axial Flow Turbines

A turbine is a prime mover or a rotary mechanical device that extracts energy from a moving fluid and generates useful work. The work done by the gas is proportional to the change of its enthalpy across the turbine. The main function of a turbine in any gas turbine engine is to drive the compressor and other accessories. In a typical jet engine, around 75% of the power produced internally is used to drive the compressor alone. In order to provide power to the compressor, a turbine has to generate as much as 50,000 hp or more for larger jet engines. One bucket or blade of a turbine will be capable of generating around 700-800 hp of power from the moving gas stream.

The working of a turbine is quite opposite to an axial flow compressor. In a compressor, the air is compressed to a high pressure whereas in the turbine it is expanded to lower pressure conditions. It must be understood that in an axial flow compressor, the fluid is made to flow against the adverse pressure gradient and is very likely to result in flow separation. To prevent this, the pressure rise through the compressor is very gradual and this certainly increases the number of stages. Hence, the pressure rise per stage is restricted to a smaller value. Therefore 10-20 stages are being used to achieve high pressure ratio in an axial flow compressor. In contrast the pressure of the fluid falls in the direction of the fluid flow and the condition in the turbine is predominantly favourable pressure gradient. In such condition, the fluid is less likely to separate when expanded drastically in the turbine. Hence, pressure expansion is quick in a turbine and this can be achieved in less number of stages as opposed to an axial flow compressor. Normally, a single stage turbine can be employed to drive even 6-7 compressor stages on the same shaft. Furthermore, in a turbojet engine, a very few number of stages are sufficient to expand the gases as the remaining gases are always expanded in the nozzle downstream. Because of these reasons, it has been observed that a well-designed turbine has a higher efficiency as compared to an axial flow compressor under normal working environment. Besides, the design process of a turbine is simpler and easier than the compressor.

Based on the nature of flow, the turbines can be broadly divided into three types namely

1. Axial flow turbines
2. Radial flow turbines
3. Mixed flow turbines

In case of axial flow turbines, the flow is in axial direction, in radial turbines the flow is in radial direction and in mixed type, both axial and radial flow occurs. Of course, the choice of turbine completely rests on the application and for jet engines, axial turbine are used as the direction of the exhaust gases in a jet engine is always required to be parallel to the engine axis. Further, in axial flow turbines we have the impulse and reaction turbines.

Description of the turbine operation

A single stage axial turbine comprises of a row of stator blades followed by a row of rotor blades very similar to the axial flow compressors. In case of a multistage turbine, it usually consists of a sequence of such stages. These sequences of alternating stator and rotor blades are covered by a shroud. The main function of a shroud is to reduce blade vibration, control of tip leakage of air passed over the blade tips. Besides these the shrouds provide resistance to the flow distortion under high loads. The schematic of an axial flow turbine stage and the shrouded turbine are shown in the figures. The use of shrouds is more common in turbines than compressors due to higher pressure ratios (hence greater leakage). The combination of stator and the shroud is often referred to as nozzle. The nozzle is always followed by rotor blade. In turbines, the length of the nozzle and the rotor blades are always increasing which is mainly done to accommodate the rapidly expanding gases while maintaining a uniform axial velocity through the stage.

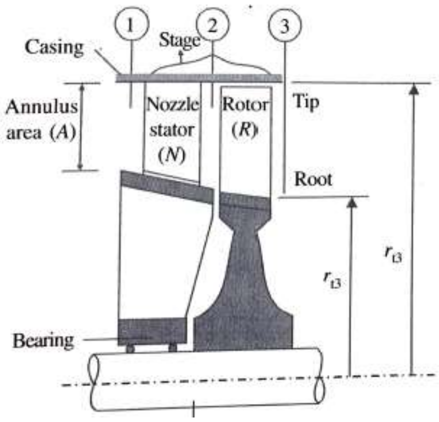


Fig: Axial flow turbine Stage

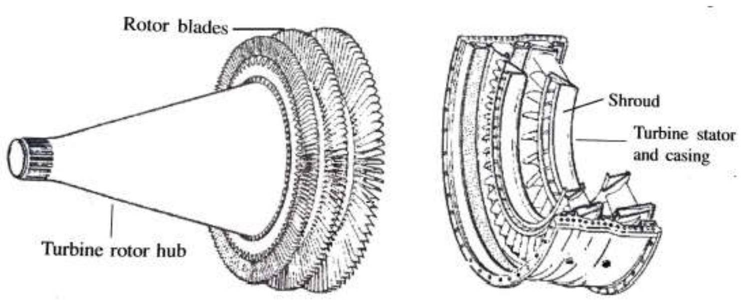
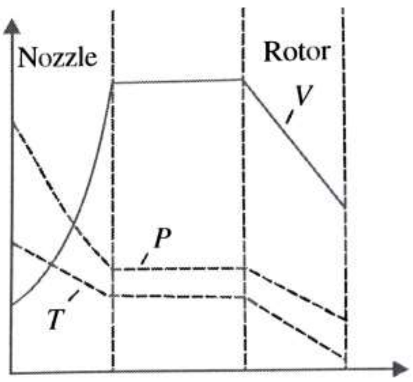


Fig: Axial flow Turbine with shroud

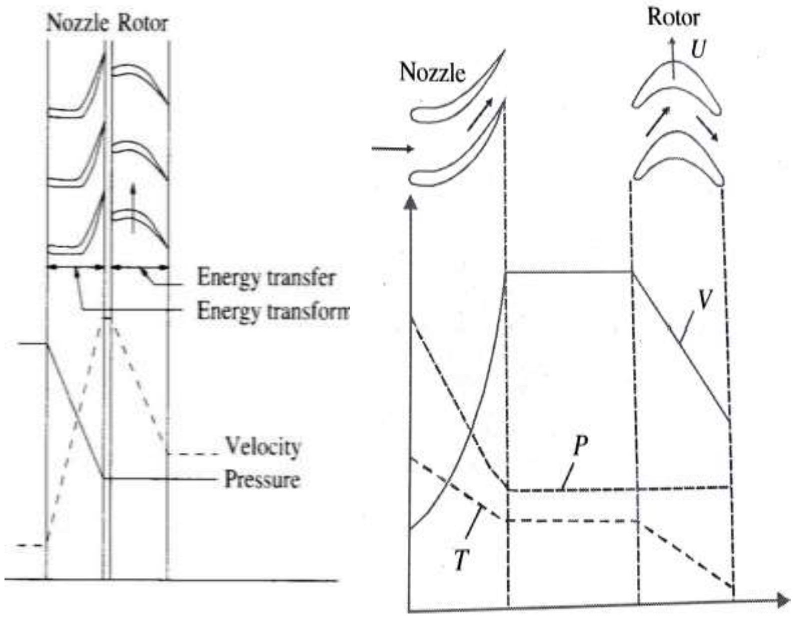
In a typical axial flow turbine, the high pressure and temperature gas from the combustion chamber is passed through the nozzles (stator) which has a converging passage, and due to this the pressure energy is converted to kinetic energy rise and a drop in static enthalpy. Next, the high velocity gases from the stator will strike the rotor blades which convert a part of the kinetic energy to shaft work through the imparted torque. In case of reaction turbines, further, expansion of the gases may occur even in the rotor part resulting in an additional increase in kinetic energy coupled with decrease in enthalpy and a part of this kinetic energy is further converted to shaft work. The variation of pressure, temperature and enthalpy across a stage is shown in the figure.



Impulse Turbine

An impulse turbine is a class of turbine in which the gas is expanded only in the stator (nozzles). A simple axial flow turbine consists of an alternating set of stationary blades followed by a set of rotating blades. The sequence of blades is exactly opposite to that of the arrangement on a compressor, where the first set of blades is the rotor blades later followed by stationary blades. Here, the stationary blades are fixed to the outer casing of the engine and the passages between the stators are similar to that of a nozzle. Next, the rotor blades are attached to the rotating spindle or the output shaft.

When the axial flow turbine is working, first the high pressure exhaust gases issuing out of the combustion chamber are passed through the stator blades. As the exhaust gases passes through the stator blade passages, the pressure energy is partially converted into velocity. Now, the high velocity exhaust gases issuing out of the nozzles strike the rotor blades. As the exhaust gases are impinged on the rotor blades, they exert an impulsive force on to the rotor blades. Since the rotor blades are free to move, upon the action of the exhaust gases, they absorb the kinetic energy from the gases and rotate. This rotational mechanical energy will be converted into electrical energy to be used for driving the compressor as well as other auxiliary components. The blades are designed such that, the exhaust gases will glide over the blades smoothly without striking and causing any pressure losses in the cycle.



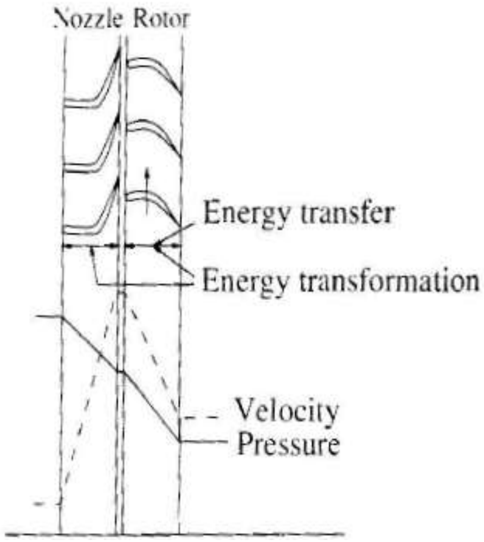
Pressure and Velocity variation in an Impulse Turbine

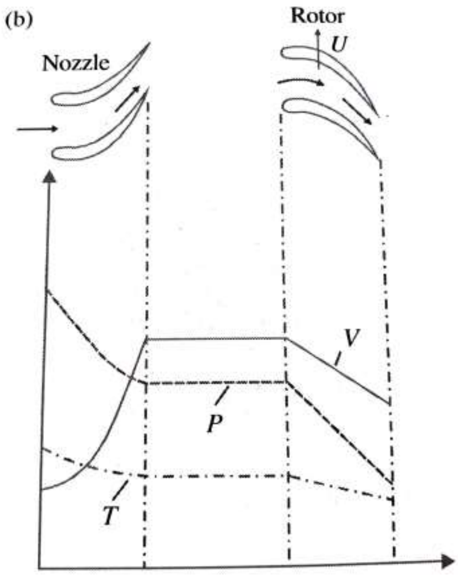
From the below figure, we can notice that, the high pressure exhaust gases with pressure Pb enters the turbine. In the stator part of the turbine, the exhaust gases undergo expansion wherein the pressure of the exhaust gases drop from Pb to Pe with a subsequent rise in velocity from vb to ve at the nozzle exit. It can be observed that the final velocity of the exhaust gases increase and will become maximum at the nozzle exit. Energy transformation occurs in the stator part of the turbine. Further, the high velocity gases strike the blades which absorb the kinetic energy of the exhaust gases. Upon absorption of kinetic energy, the velocity of the gases decreases from ve to vf at the rotor exit and in due course it will cause the bade to rotate. Finally, in the rotor part, the process of energy transfer occurs, thus generating energy.

Reaction Turbine

A reaction turbine is one in which the expansion of gases occur in both the stator and rotor part of the turbine. In reaction turbines, both the stator as well as the rotor is convergent in nature which causes the process of energy transformation in stator and rotor, thus making the reaction turbine more efficient than the impulse turbines.

During the working of a reaction turbine, the high pressure exhaust gas from the combustion chamber enters the first row of stator blades. In the stator blades, the velocity of the exhaust gases are increased with a simultaneous decrease in pressure energy i.e., energy transformation occurs. The exhaust gas leaves the stator with a specific velocity and enters the rotor stage. Once entering the first rotor stage, the exhaust gases will see the rotor passages as a convergent duct. The change in area from the inlet to outlet produces an increase in relative velocity with a accompanying pressure drop. The acceleration of the gases in the rotor stage of the turbine generates a reaction force similar to the one produced on an airplane wing. It is from this feature of the reaction turbine its name has been derived. The direction of the reaction force is always perpendicular to the blade similar to the lift on an airfoil.





Aircraft Nozzles

A nozzle is relatively simple device with a varying cross section and use to direct or modify the flow as it exits the chamber. In an aircraft application it is mainly used to convert the pressure energy of the fluid into kinetic energy and eject fluid in a coherent stream into a surrounding medium. Nozzles are frequently used to control the rate of flow, speed, direction, mass, shape, and/or the pressure of the stream that emerges from them. Propelling nozzles accelerate the available gas to subsonic, sonic, or supersonic velocities depending on the power setting of the engine, their internal shape and the pressures at entry to, and exit from, the nozzle. The main functions of any nozzle are:

1. Accelerate the flow to a high velocity with minimum total pressure loss
2. Match exit and atmospheric pressure as closely as possible

3. Permit afterburner operation without affecting main engine operation-this function requires a variable area nozzle
4. Allow for cooling of walls
5. Mix core and by pass streams of turbofan if necessary
6. Allow for thrust reversing
7. Suppress jet noise and infrared radiation if desired
8. Thrust vector control should be permitted

There are different types of nozzles used on aircrafts:

1. Convergent Nozzle
2. Convergent-Divergent nozzle or Con-Di nozzle or De-Laval nozzle

Convergent Nozzle

A convergent nozzle is a simple convergent duct as shown in the below figure. This type of nozzles is mainly used when the pressure ratio (P_b/P_o) is less (less than about 4). Convergent nozzles have a fixed exit area and can accelerate the flow only up to sonic (M=1) condition. The convergent nozzle has been generally used in engines for subsonic aircrafts.

Fig : Convergent Nozzle

Flow through a Converging Nozzle

Consider a converging nozzle connected to a reservoir where stagnation conditions prevail, P = P_0, T = T_0, U = 0(Velocity). By definition reservoirs are such that no matter how much the fluid flows out of them, the conditions in them do not change. In other words, pressure, temperature, density etc. remain the same always.

Back Pressure- Pressure level P_b at the exit of the nozzle is referred to as the Back Pressure and it is this pressure that determines the flow in the nozzle. It is also defined as the pressure of the fluid which surrounds the nozzle. Let us now study how the flow responds to changes in Back Pressure.

When the Back Pressure, P_b is equal to the reservoir pressure, P_0, there is no flow in the nozzle. This is condition (1) in Fig 1. Let us reduce p_b slightly to P_2 (condition (2) in the Figure). Now a flow is induced in the nozzle due to the pressure differences. For relatively high values of P_b, the flow is subsonic throughout. A further reduction in Back Pressure results in still a subsonic flow, but of a higher Mach number at the exit (condition (3)). Note that the mass flow rate increases. As P_b is reduced we have an increased Mach number at the exit along with an increased mass flow rate. If the back pressure is lowered further then at a particular back pressure value the flow reaches sonic conditions at the throat of the nozzle (4). Usually this occurs when the back pressure ratio reaches a value which is given below.

$$\frac{P_b}{P_o} = 0.5283$$

When the back pressure is further lowered, the Mach number at the exit tries to increase. It demands an increased mass flow from the reservoir. But as the condition at the exit is sonic, signals do not propagate upstream. The Reservoir is unaware of the conditions downstream and it does not send any more mass flow. Consequently the flow pattern remains unchanged in the nozzle. Any adjustment to the Back Pressure takes place outside of the nozzle. The nozzle is now said to be choked. The mass flow rate through the nozzle has reached its maximum possible value, choked value. From Fig 1 we see that there is an increase in mass flow rate only till choking condition (4) is reached. Thereafter mass flow rate remains constant.

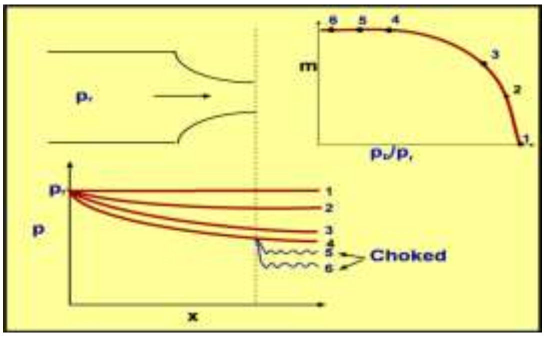


Fig : Flow through a Converging Nozzle

When the back pressure ratio is large enough, the flow within the entire device will be subsonic and isentropic. When the back pressure ratio reaches a critical value, the flow will become choked with subsonic flow in the converging section, sonic flow at the throat, and subsonic flow in the diverging section. When the condition at the throat is sonic (M=1) then the flow is said to be sonic. This condition of the nozzle is said to be chocked. **Choked flow** is a limiting condition which occurs when the mass flow rate will not increase with a further decrease in the downstream pressure environment while upstream pressure is fixed.

Convergent-Divergent Nozzle or Con-Di Nozzle or De-Laval Nozzle

Convergent-Divergent nozzle was invented by Carl de Laval towards the end of the l9th century and is thus often referred to as the 'de Laval' nozzle. A convergent-Divergent nozzle has two sections i) Convergent Duct and ii) Divergent Duct. The section of the nozzle where the cross-sectional area is minimum is called the throat of the nozzle. Most Con-Di nozzles used on aircrafts are not simple ducts. They incorporate variable geometry and other aerodynamic features. These nozzles are mainly used when the pressure ratios are more than 6 (P_b/P_o >6). These nozzles have the ability to accelerate the flow to supersonic speeds and hence they are a characteristic feature of all supersonic aircrafts. Also if the engine incorporates an afterburner, a convergent-divergent nozzle will be used. In such cases the throat of the nozzle will be made in such a way that the operating conditions of the engine upstream of the afterburner remain unchanged. Further, the exit area must be varied to match the flow conditions in order to produce the maximum thrust.

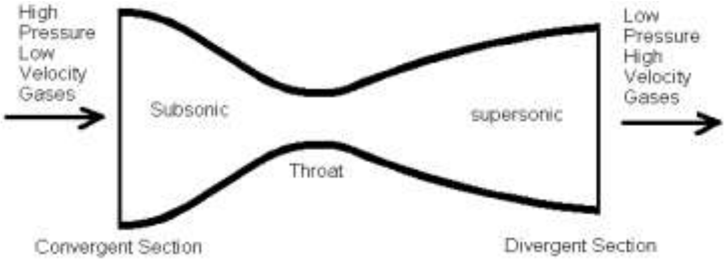
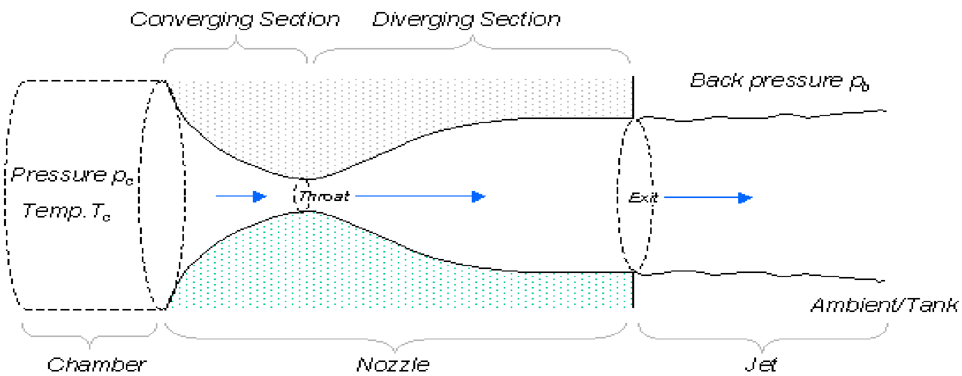


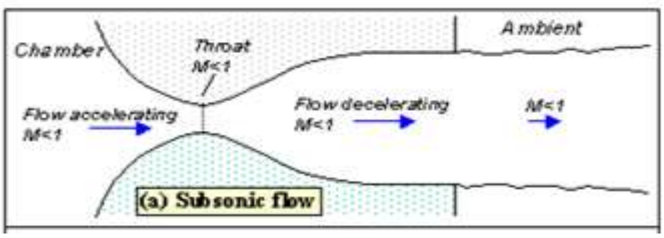
Fig: Convergent-Divergent Nozzle

The usual configuration for a converging diverging (CD) nozzle is shown in the above figure. Gas flows through the nozzle from a region of high pressure (usually referred to as the chamber) to one of low pressure (referred to as the ambient or tank). The chamber is usually big enough so that any flow velocities here are negligible. The pressure here is denoted by the symbol P_0. Gas flows from the chamber into the converging portion of the nozzle, past the throat, through the diverging portion and then exhausts into the ambient as a jet. The pressure of the ambient is referred to as the 'back pressure' and given the symbol P_b.

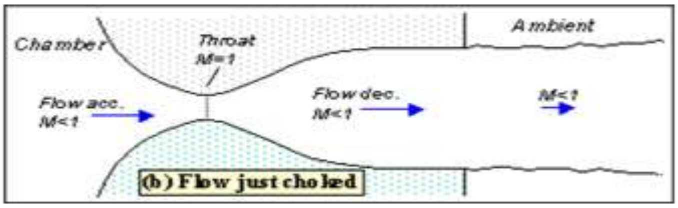
Let us imagine a convergent-divergent nozzle to be connected to two air cylinders. Cylinder A contains air at high pressure, and takes the place of the chamber. The CD nozzle exhausts this air into cylinder B, which takes the place of the tank. Imagine you are controlling the pressure in cylinder B i.e., reducing the back pressure p_b, and measuring the resulting mass flow rate through the nozzle.



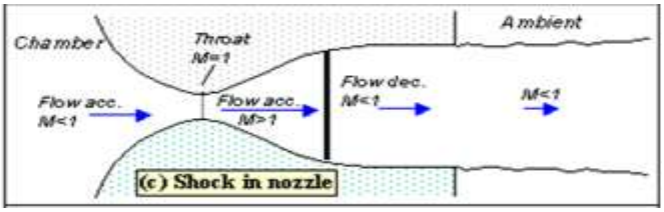
Initially when the back pressure is same as the chamber pressure, there won't be any pressure differences induced and the flow remains stagnant (No movement). When the back pressure is reduced slightly then flow accelerates out of the chamber through the converging section, reaching its maximum (subsonic) speed at the throat. The flow then decelerates through the diverging section and exhausts into the ambient as a subsonic jet. Lowering the back pressure in this state increases the flow speed everywhere in the nozzle. This condition 3a is depicted in the below fig 3.



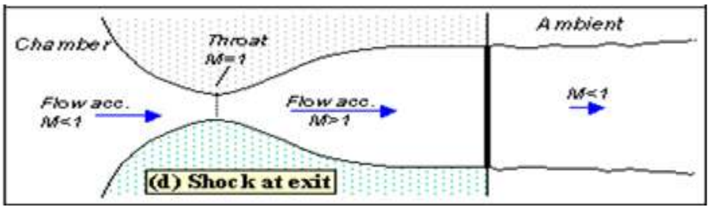
Lower it far enough and we eventually get to the situation shown in figure 3b. The flow pattern is exactly the same as in subsonic flow, except that the flow speed at the throat has just reached M=1. Flow through the nozzle is now choked since further reductions in the back pressure can't move the point of M=1 away from the throat. However, the flow pattern in the diverging section does change as you lower the back pressure further.



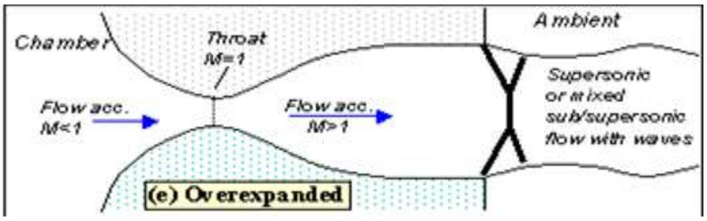
As p_b is lowered below that needed to just choke the flow a region of supersonic flow forms just downstream of the throat (Fig 3c). Unlike a subsonic flow, the supersonic flow accelerates as the area gets bigger. This region of supersonic acceleration is terminated by a normal shock wave. The shock wave produces a near-instantaneous deceleration of the flow to subsonic speed. This subsonic flow then decelerates through the remainder of the diverging section and exhausts as a subsonic jet. In this regime if you lower or raise the back pressure you increase or decrease the length of supersonic flow in the diverging section before the shock wave.



If the back pressure is continuously lowered, then the shock wave tends to move towards the exit of the nozzle and sits at the exit plane of the nozzle (Fig 3d). Here a very long region of acceleration (the entire nozzle length) can be achieved. In this case the flow speed will be supersonic upto the shock at the nozzle exit. However, after the shock the flow will still become again subsonic.



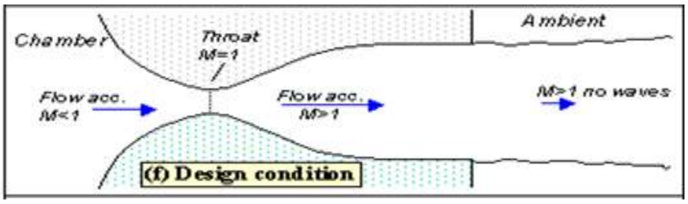
Lowering the back pressure further causes the shock to bend out into the jet (figure 3e), and a complex pattern of shocks and reflections is set up in the jet which will now involve a mixture of subsonic and supersonic flow, or (if the back pressure is low enough) just supersonic flow. Because the shock is no longer perpendicular to the flow near the nozzle walls, it deflects it inward as it leaves the exit producing an initially contracting jet. We refer to this as **over expanded flow** because in this case the pressure at the nozzle exit is lower than that in the ambient (the back pressure)- i.e. the flow has been expanded by the nozzle too much.



Over expanded Nozzle

- Fluid exits at pressure lower than the atmospheric pressure
- This owes to an exit area too large for optimum

A further lowering of the back pressure changes and weakens the wave pattern in the jet. Eventually we will have lowered the back pressure enough so that it is now equal to the pressure at the nozzle exit. In this case, the waves in the jet disappear altogether (figure 3f), and the jet will be uniformly supersonic. This situation, since it is often desirable, is referred to as the "**design condition**".



Finally, if we lower the back pressure even further we will create a new imbalance between the exit and back pressures (exit pressure greater than back pressure), figure 3g. In this situation **(called 'Underexpanded')** what we call expansion waves (that produce gradual turning and acceleration in the jet) form at the nozzle exit, initially turning the flow at the jet edges outward in a plume and setting up a different type of complex wave pattern.

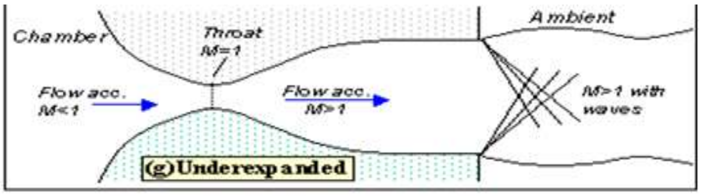


Figure 3. Flow patterns

Underexpanded Nozzle

- Discharges fluid at an exit pressure greater than the external pressure
- This owes to the exit area being too small for an optimum area ratio
- The expansion of the fluid is incomplete
- Further expansion happens outside of the nozzle
- Nozzle exit pressure is greater than local atmospheric pressure

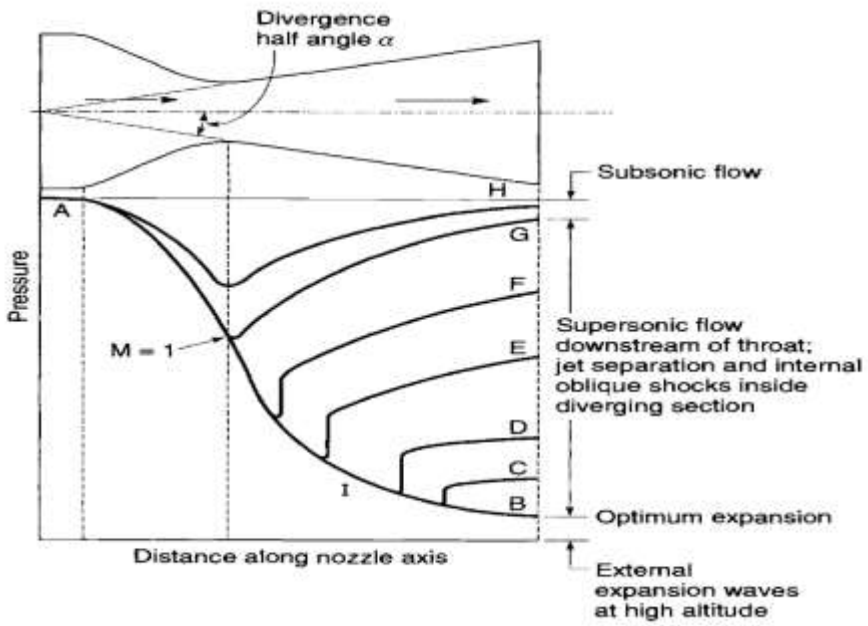


Fig: Distribution of pressures in a C-D nozzle for different flow conditions

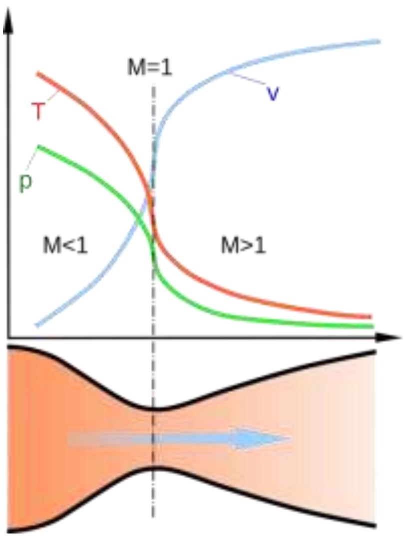


Fig : Pressure, Velocity & Temperature Variation through a C-D Nozzle